Introduction to Mental Models

Based on Topics in Sakal-Column

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Introduction

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Introduction to Mental Models

- Mental models are frameworks to simplify complex decision-making.
- Popularized by Charlie Munger and Shane Parrish.
- Originates from cognitive psychology and systems theory.
- Helps in investing (e.g., Munger's latticework), policymaking, and life choices.
- E.g., Ratan Tata applying mental models in Tata Nano's design.
- Be aware of over-relying on one model; use a toolbox approach.

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What Are Mental Models?

- Mental Models help us understand and interpret the world.
- They shape how we think, what we notice, and what we ignore.
- Mental Models simplify complexity and highlight relevance.
- They guide reasoning, decision-making, and perception.
- Every thought process relies on some form of Mental Model.

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Mental Models: From Data to Knowledge

- Information = raw data.
- Thinking = organizing information into structure.
- Knowledge = structured information with meaning.
- Mental Models are structured knowledge frameworks.
- They convert data into insight by imposing structure and meaning.
- Information + Thinking = Knowledge

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Why Do We Have Mental Models?

- Brains evolved to predict environmental changes and increase survival odds.
- Mental Models are tools the brain uses to make accurate predictions.
- Decision-making is based on predictions, consciously or subconsciously.
- Mental Models help assign meaning to experiences and emotions.
- We use them in both logical decisions and emotional interpretations.
- Even basic reactions (e.g. Fight or Flight) are driven by primal models.
- Mental Models fill gaps in uncertainty to help us make sense of the world.

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Mental Models as Predictive Tools

- All decisions are bets on Mental Model-based predictions.
- Models like 80/20 or Second Order Thinking predict system outcomes.
- Mental Models help evaluate probabilities in uncertain environments.
- They convert external stimuli into internal meaning and action.
- Interpretation of social cues often uses layered Mental Models.
- Miscalibrated models can lead to incorrect assumptions and behavior.
- Models operate both reactively (instinct) and reflectively (consciously).
- They underpin both logical reasoning and emotional reactions.

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Mental Models

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Mental Models at a Glance

- **Reasoning & Thinking**: First Principles, Inversion, Second-Order Thinking, Thought Experiment, Probabilistic Thinking, Falsifiability
- **Decision-Making**: Opportunity Cost, Sunk Cost Fallacy, Pareto Principle, Circle of Competence, Occam's Razor
- **Psychology & Bias**: Cognitive Biases, Confirmation Bias, Dunning-Kruger Effect, Availability Heuristic, Survivorship Bias
- **Systems & Physics**: Systems Thinking, Feedback Loops, Inertia, Thermodynamics, Leverage, Activation Energy, Friction, Velocity
- **Human & Social**: Reciprocity, Skin in the Game, Incentives, Tragedy of the Commons, Cooperation, Hanlon's Razor
- **Biology & Ecology**: Evolution, Ecosystems, Niches, Adaptation, Self-Preservation, Catalysts, Alloying

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First Principles Thinking

- Break problems down to their fundamental truths.
- Originates from Aristotle and physics.
- Used in startups, e.g., Ola dissecting cab logistics vs. copying taxi systems.
- In cooking, understanding base ingredients vs. recipes.
- E.g., ISRO's cost-effective Mars mission using fundamental physics and constraints.
- Beware of ignoring proven heuristics and reinventing unnecessarily.

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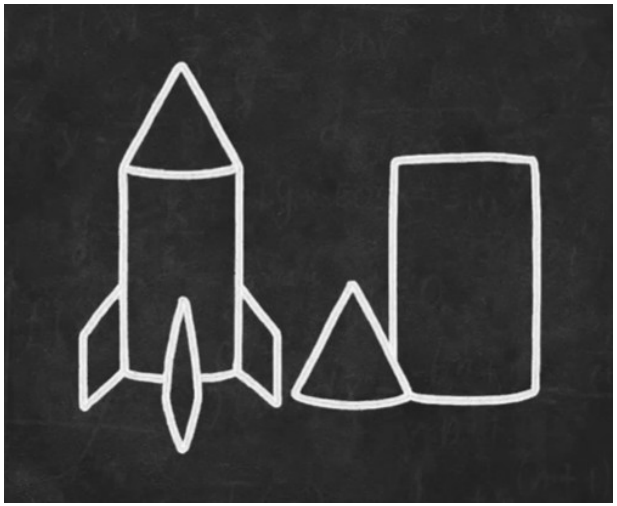
First Principles Thinking

- **Definition**: Break a problem down to its most fundamental, indisputable truths, then reason upward from scratch rather than relying on analogy or convention.
- **How to Apply**: List every assumption you hold about the problem. Challenge each one by asking "Is this definitely true?" Discard inherited wisdom that cannot be verified, and rebuild your solution only from what survives.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Take a belief you hold about your work or life and ask "Why?" five times in a row until you hit bedrock fact.

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Inversion

- Thinking in reverse , "what would cause failure?"
- Inspired by Carl Jacobi's "invert, always invert."
- In finance: prevent loss before seeking gain.
- In daily life: avoiding bad health rather than only pursuing good.
- E.g., B-schools teaching case studies of failed Indian startups.
- Avoid over-focusing on negatives; it's a tool, not a worldview.

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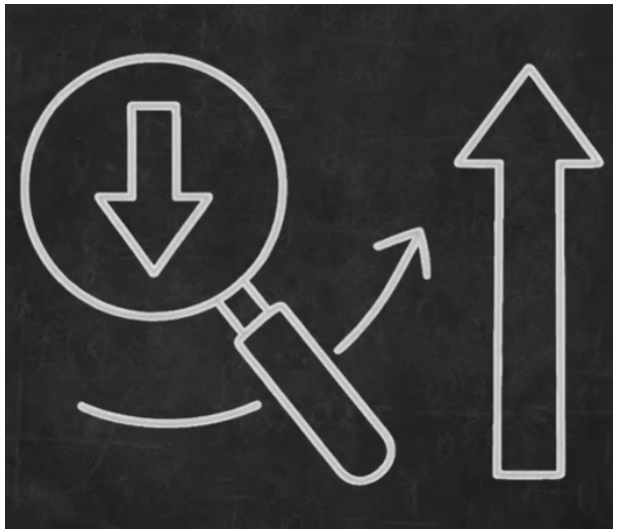
Inversion

- **Definition**: Instead of asking how to achieve a goal, ask what would guarantee failure, then avoid those things.
- **How to Apply**: Define the desired outcome clearly. Brainstorm every action, habit, or condition that would guarantee you never reach it. Eliminate or guard against those failure modes rather than only chasing success.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Before your next project kick-off, write a "pre-mortem": assume it failed completely and list the causes.

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Opportunity Cost

- Cost of the next best alternative foregone.
- Core principle in economics since classical era.
- Choosing civil services over entrepreneurship , missed upside.
- In Indian agriculture: water for paddy vs. cash crops.
- E.g., investor choosing fixed deposit vs. equity.
- Don't ignore hidden or non-financial costs; measure holistically.

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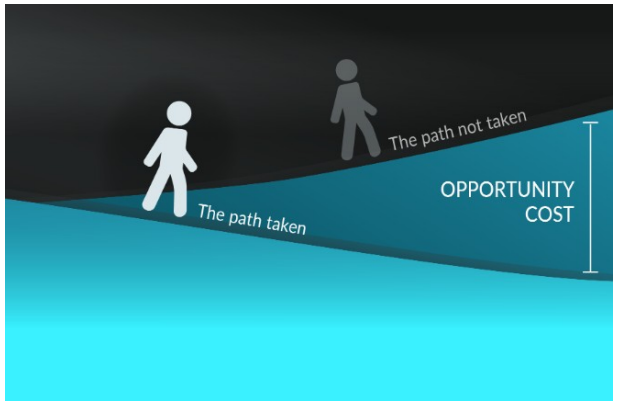
Opportunity Cost

- **Definition**: Every choice you make implicitly rejects every alternative, the value of the best foregone alternative is your true cost.
- **How to Apply**: Before committing to any decision, explicitly name the next-best alternative. Weigh the value of what you gain against the value of what you give up, not just the monetary or time price.



(Ref: "Model Thinkers" - David Green)

Try This: Before your next big "yes," write down the single best thing you are saying "no" to and decide consciously.

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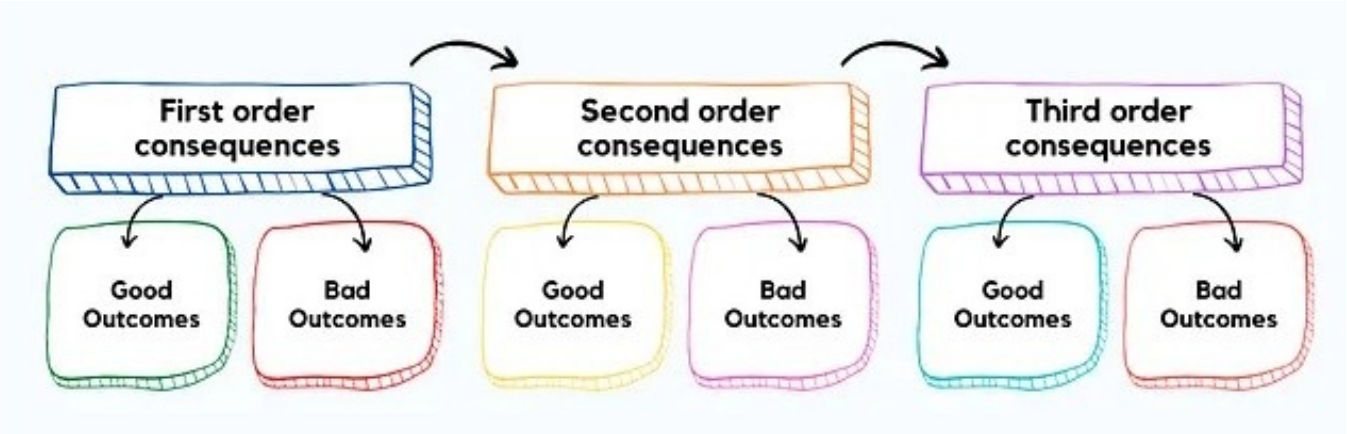
Second-Order Thinking

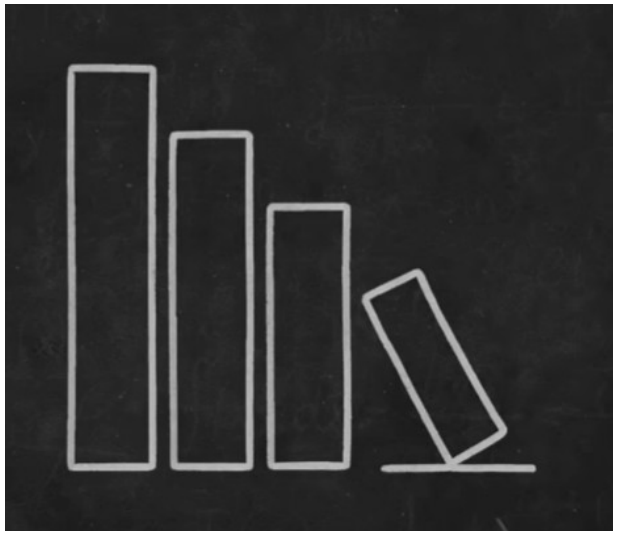
- Anticipating consequences of consequences.
- Key to systems thinking; used by top strategists.
- In politics: free electricity → higher consumption → grid stress.
- In business: discounts → loss leaders → brand erosion.
- E.g., UPI adoption → cashless economy → surveillance concerns.
- Avoid paralysis by analysis; balance depth with action.



(Ref: The Art Of Thinking Clearly - Swabhav Tech Labs)

Second-Order Thinking

- **Definition**: Think beyond the immediate consequence of a decision to the downstream effects it will trigger in turn.
- **How to Apply**: Map your decision three steps deep: "If I do X, then Y happens, and if Y happens, then Z follows." Evaluate whether Z, not just Y, is what you actually want.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: For your next important decision, write out the chain: Action $\rightarrow$ 1st effect $\rightarrow$ 2nd effect $\rightarrow$ 3rd effect.

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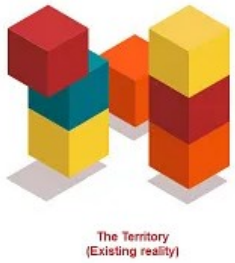
The Map is Not the Territory

- Models are simplifications , not reality itself.
- Coined by Alfred Korzybski, popular in systems science.
- GDP $\neq$ actual prosperity; rankings $\neq$ real competence.
- E.g., exam scores $\neq$ intelligence; IRCTC waitlist $\neq$ travel reality.
- E.g., BPL card $\neq$ real poverty in rural India.
- Don't confuse labels or proxies with full understanding.



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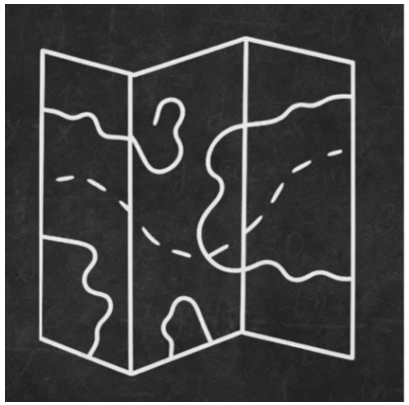
The Map is Not the Territory

- **Definition**: Any model, plan, or description of reality is a simplification, the real world always contains more detail and surprise than the map suggests.
- **How to Apply**: Treat your plans and mental models as useful approximations, not ground truth. Actively seek out what your model omits. Update your map quickly when reality contradicts it rather than defending the map.



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Try This: Identify one assumption in a current plan and go verify it directly in the real world this week.

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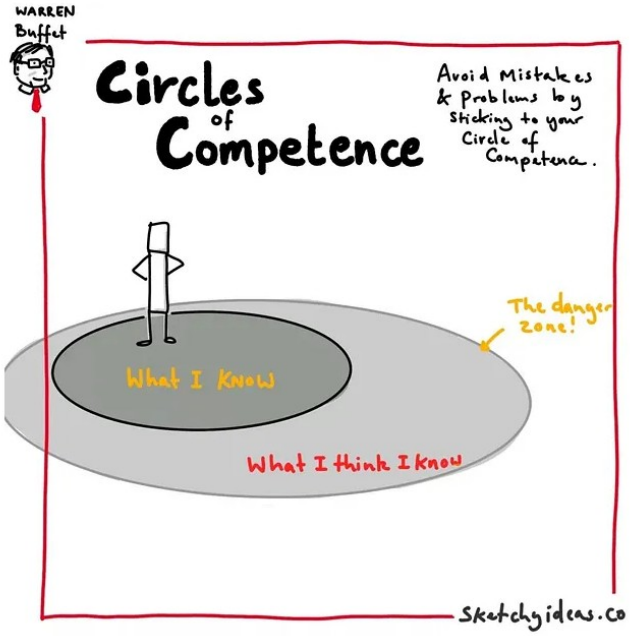
Circle of Competence

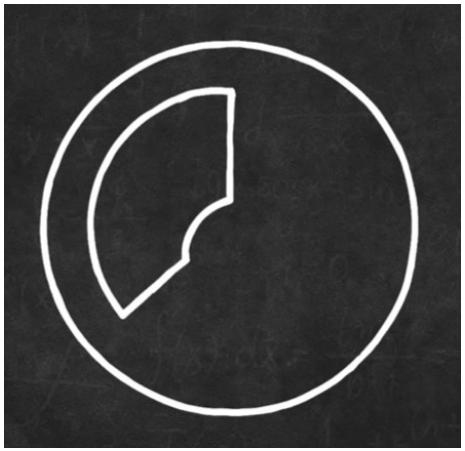
- Know what you know , and what you don't.
- Warren Buffett's key idea; deeply tied to humility.
- Investors stick to known sectors , pharma vs. tech.
- E.g., Indian cricket team sticking to seamers on green pitches.
- E.g., Byju's scaling fast in edtech, failed in physical schools.
- Don't make your circle too small , be open to learning.



Circles Of Competence (Image Credits: sketchyideas)

Circle of Competence

- **Definition**: Stick to decisions within the domain you genuinely understand deeply; know and respect the boundary of your expertise.
- **How to Apply**: Draw three honest circles: what you know deeply (operate here), what you are learning (expand carefully), what is outside your knowledge (defer or avoid). Before taking action, ask: which circle is this in?



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Write down your three genuine areas of deep expertise and explicitly label your next big decision as inside or outside them.

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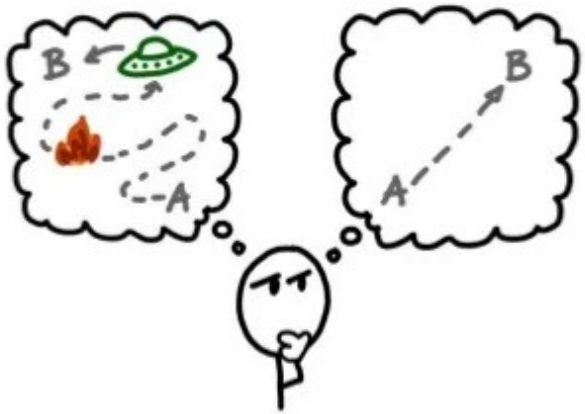
Ockham's Razor

- Simplest explanation is often correct.
- Named after William of Ockham, 14th century.
- Used in diagnosis, journalism, and science.
- E.g., missing train due to traffic vs. conspiracy by cabbie.
- E.g., Aadhaar leaks more likely due to mismanagement than espionage.
- Beware of oversimplification; simple $\neq$ accurate.



"When faced with two equally good hypotheses, always choose the simpler."

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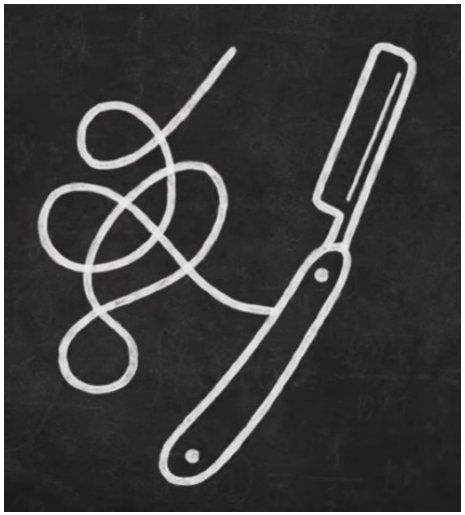
Ockham's Razor

- **Definition**: When two explanations fit the facts equally well, prefer the simpler one, complexity is not a feature, it is a cost.
- **How to Apply**: Start with the simplest possible explanation or solution. Add complexity only if the simple version demonstrably fails to explain the evidence or solve the problem.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Look at a complex solution you are building and ask: what is the minimum version that still works?

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The Lindy Effect

- The longer something survives, the longer it's likely to.
- Popularized by Nassim Taleb; derived from theater.
- Sanskrit, Ayurveda , resilient knowledge in Indian context.
- Brands like Amul, Tata indicate trust through time.
- E.g., epics like Mahabharata enduring cultural relevance.
- New $\neq$ better; but old $\neq$ always useful , evaluate critically.

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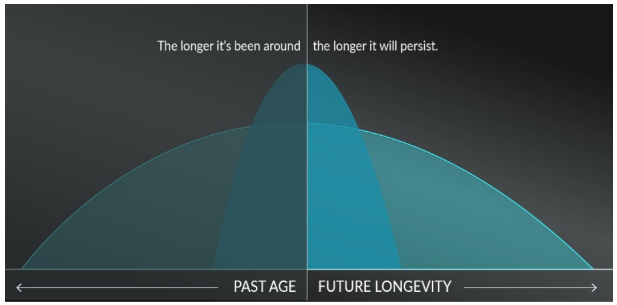
The Lindy Effect

- **Definition**: The longer something non-perishable has already survived, the longer it is likely to continue to survive in the future.
- **How to Apply**: When evaluating ideas, technologies, or practices, favour those with long track records over new entrants, especially in high-stakes domains. A book read for 500 years will likely be read for 500 more.



(Ref: "Model Thinkers" - Albert Goldman)

Try This: Before adopting a new tool or methodology, ask how old the underlying principle is, not just the product.

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Sunk Cost Fallacy

- Continuing due to past investment, not future value.
- Studied in behavioral economics.
- Government projects continued despite low returns.
- E.g., B-school student finishing MBA despite hating it.
- E.g., personal relationships carried forward "because of time invested."
- Don't throw good money (or time) after bad.

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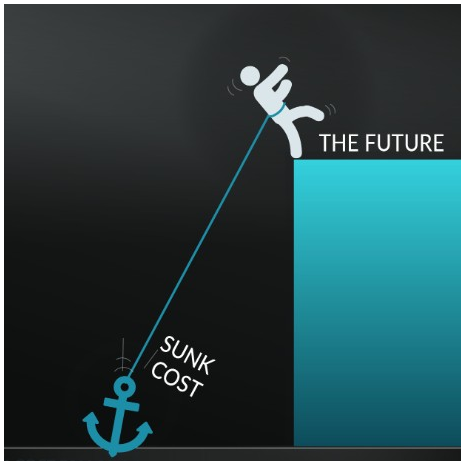
Sunk Cost Fallacy

- **Definition**: Past investment, time, money, or effort already spent, is gone regardless of what you do next and should never drive future decisions.
- **How to Apply**: Separate what you have spent from what you stand to gain going forward. Ask: "If I had not already invested in this, would I choose to start it today?" If no, stop.



(Ref: "Model Thinkers")

Try This: List one project you are continuing mainly because of what you have already put in, then honestly evaluate its future value.

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The Pareto Principle (80/20 Rule)

- 80
- Discovered by Vilfredo Pareto in economics.
- E.g., 20
- In exams, 20
- E.g., Indian IT firms rely heavily on top 20
- Avoid ignoring the remaining 80

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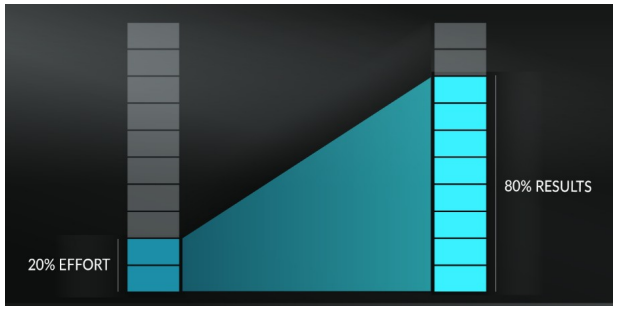
The Pareto Principle (80/20 Rule)

- **Definition**: Roughly 80% of outcomes come from 20% of causes, a small number of inputs drive the vast majority of results.
- **How to Apply**: Track your activities for one week and identify which 20% produce 80% of your meaningful results. Ruthlessly protect time for those activities and minimise or delegate the rest.



(Ref: "Model Thinkers" - Joseph Juran)

Try This: List your top 10 tasks and circle the two that drive most of your results, then schedule those first tomorrow.

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Cognitive Biases

- Systematic deviations from rational thinking.
- Studied extensively by Kahneman and Tversky.
- Media bias, confirmation bias, anchoring , common in investing and politics.
- E.g., brand loyalty despite better alternatives.
- Indian voters influenced by recency bias during elections.
- Learn biases to spot them, not to feel superior.

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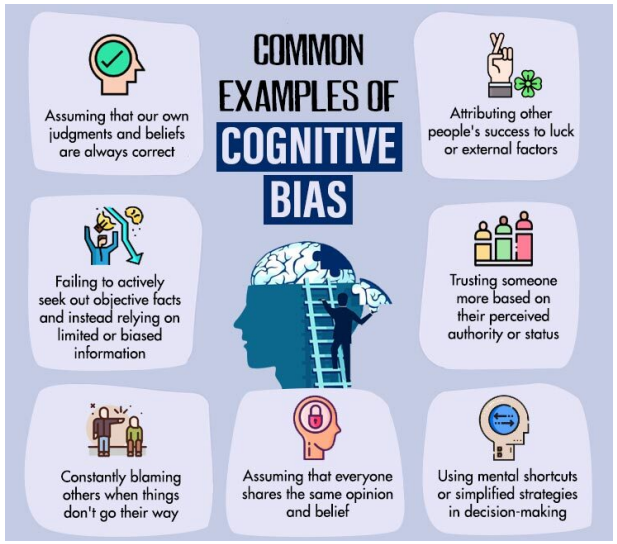
Cognitive Biases

- **Definition**: Systematic errors in thinking that cause us to deviate from rational judgment, usually without awareness.
- **How to Apply**: Learn the most common biases (anchoring, availability, confirmation, etc.) and build check-in questions into important decisions. Slow down deliberate thinking when stakes are high.



(Ref: "Cognitive Biases" - Mind dot Help)

Try This: Before your next major decision, name one bias most likely to distort your thinking and design a simple counter-check.

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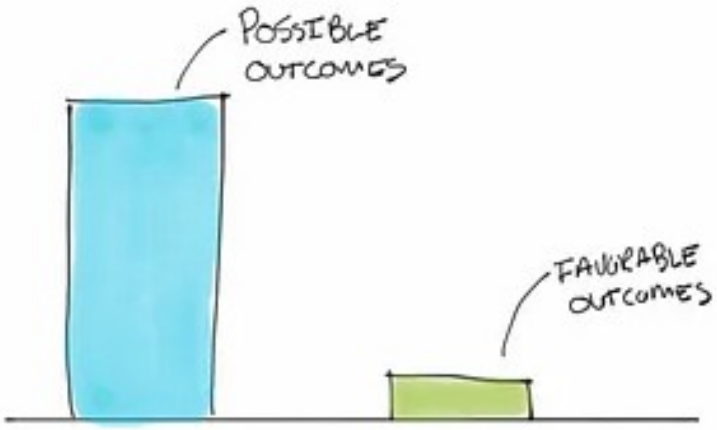
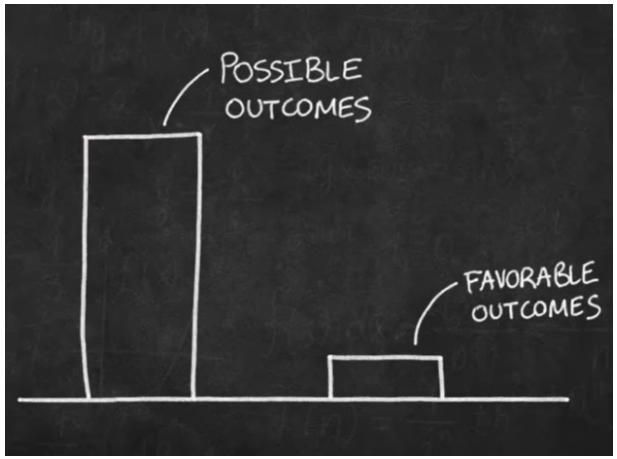
Probabilistic Thinking

- Thinking in terms of likelihoods, not certainties.
- Core to Bayesian reasoning and decision theory.
- E.g., cricket match predictions, monsoon forecasts.
- Stock investors use expected value calculations.
- E.g., Indian RTOs plan road design using accident probabilities.
- Avoid overconfidence; even 90



Probabilistic Thinking

- **Definition**: Think in likelihoods and ranges rather than certainties, assign probabilities to outcomes rather than treating one scenario as inevitable.
- **How to Apply**: For any forecast, identify the best case, worst case, and most likely case with rough probability weights. Update your probabilities as new evidence arrives rather than locking in an early view.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: For your next big decision, explicitly write: "I think there is a __% chance this works, __% it partially works, __% it fails."

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The Dunning-Kruger Effect

- Incompetent people overestimate their competence.
- Coined by psychologists Dunning and Kruger.
- E.g., novice trader in India believing they're stock experts post one gain.
- Political debates full of loud but uninformed opinions.
- E.g., early success of startups can lead to overexpansion.
- Competence includes knowing limits; humility is key.

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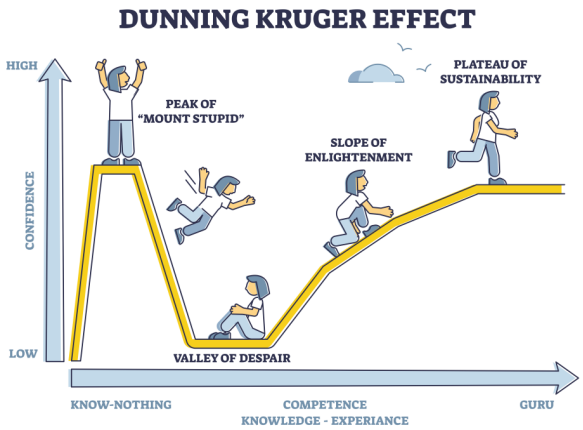
The Dunning-Kruger Effect

- **Definition**: People with limited knowledge in a domain tend to overestimate their competence; true experts tend to underestimate it.
- **How to Apply**: Be most sceptical of your own confidence when you are new to a subject. Seek expert review of your work. Interpret high certainty as a signal to double-check, not a green light.



(Ref: "Understanding the Dunning-Kruger Effect and Its Impact on Leadership" - Curt Archambault)

Try This: In a domain where you feel very confident, find one expert who disagrees with you and genuinely try to understand their view.

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Hanlon's Razor

- Never attribute to malice what can be explained by stupidity.
- Popular in management and conflict resolution.
- E.g., government inefficiency $\neq$ conspiracy.
- Indian Railways delays often due to process, not sabotage.
- In family arguments, ignorance may explain behavior better than bad intent.
- Beware of excusing true malice , discern with care.

Hanlon's Razor

- **Definition**: Never attribute to malice what can be adequately explained by incompetence, ignorance, or accident.
- **How to Apply**: When someone's action harms or frustrates you, consider the most charitable explanation first. Ask whether the person had the knowledge, intention, and motive to act maliciously before concluding they did.

(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: The next time you feel wronged, write two alternative explanations, one malicious, one accidental, and investigate before deciding which fits.

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Systems Thinking

- Viewing elements as parts of interrelated wholes.
- Rooted in cybernetics and ecology.
- E.g., Indian river pollution due to upstream waste + politics + habits.
- Public health influenced by education, nutrition, and social norms.
- E.g., traffic = vehicles + roads + behavior + enforcement.
- Risk: complexity can obscure actionable insight , avoid analysis paralysis.

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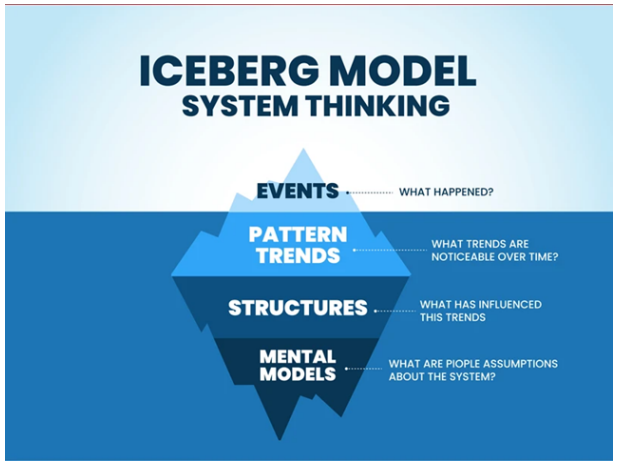
Systems Thinking

- **Definition**: See the world as interconnected systems with feedback loops rather than as isolated chains of cause and effect.
- **How to Apply**: Map the key elements of a problem and draw the relationships between them, including feedback loops. Look for delays, reinforcing loops that amplify, and balancing loops that dampen, then intervene at high-leverage points.



(Ref: "Systems Thinking: A Holistic Approach to Solving Complex Problems" - 6 Sigma)

Try This: Draw a simple diagram of your biggest current problem showing at least three elements and how they influence each other.

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The Availability Heuristic

- People judge likelihood by what comes easily to mind.
- From cognitive psychology.
- After plane crash news, people avoid flying.
- E.g., fear of crime rising due to sensational Indian news channels.
- COVID panic in India driven by social media images.
- Recognize bias: what's vivid $\neq$ what's common.

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The Availability Heuristic

- **Definition**: We judge the likelihood of events by how easily examples come to mind, causing vivid or recent events to feel far more probable than they are.
- **How to Apply**: When assessing risk or frequency, deliberately seek out base-rate statistics rather than relying on memorable anecdotes. Ask: "What does the data say?" before asking: "What do I recall?"

(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one risk you worry about disproportionately and look up its actual statistical probability.

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Skin in the Game

- Decision-makers must share risks of their actions.
- Advocated by Nassim Taleb.
- E.g., Indian politicians rarely affected by laws they make.
- Business owners risking personal capital vs. salaried CEOs.
- E.g., doctors prescribing tests vs. family treating conservatively.
- Don't assume skin always = alignment; people may still act irrationally.

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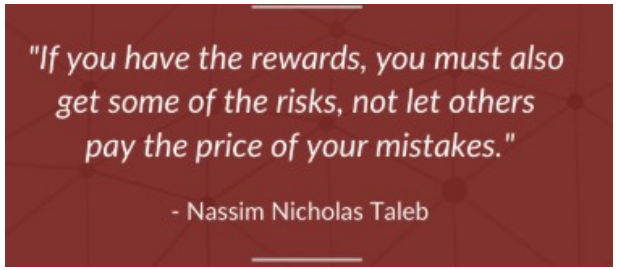
Skin in the Game

- **Definition**: Decision-makers should bear real consequences, upside and downside, for the outcomes of their choices.
- **How to Apply**: Align incentives so that the people making recommendations also share in the results. When seeking advice, prefer those who have personal stakes in the outcome over pure observers.



(Ref: "Book Summary – Skin in the Game: The Hidden Asymmetries in Daily Life" - Reading Graphics)

Try This: Before taking advice, ask: "What does this person lose if they are wrong?"

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Confirmation Bias

- Tendency to seek info that confirms preconceptions.
- Found in psychology and behavioral economics.
- E.g., political debates in India , media consumed selectively.
- Investors seek news that supports their portfolio.
- E.g., religious or caste-based beliefs reinforced via social networks.
- Actively seek disconfirming evidence for better clarity.

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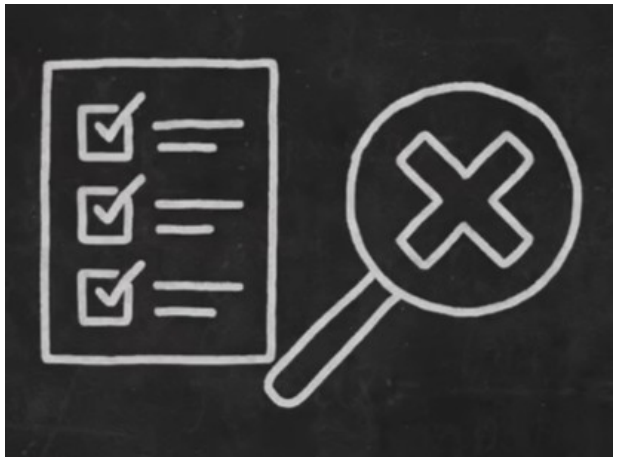
Confirmation Bias

- **Definition**: We unconsciously seek, interpret, and remember information that confirms what we already believe.
- **How to Apply**: Actively search for evidence that contradicts your current view. Assign a "devil's advocate" role in group decisions. Weight disconfirming data at least as heavily as confirming data.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Read one article or talk to one person who holds the opposite view to yours on an important topic before finalising your position.

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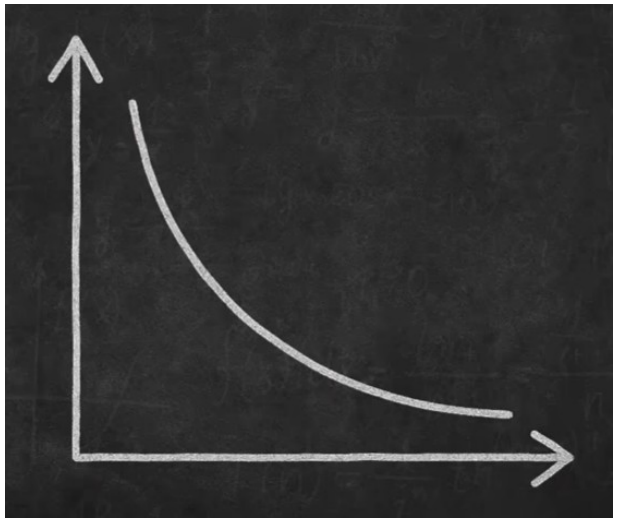
The Law of Diminishing Returns

- Each additional input yields less output after a point.
- Core to economics and productivity.
- E.g., studying 10 hrs vs. 20 hrs doesn't double marks.
- Government subsidies show lower impact over time.
- E.g., fertilizer overuse harming Indian soils.
- Be aware of optimal input level , don't push endlessly.

YHK

The Law of Diminishing Returns

- **Definition**: Beyond a certain point, adding more input yields progressively smaller increases in output, and eventually none at all.
- **How to Apply**: Find the inflection point in any effort, the moment when additional time, money, or resources stop yielding meaningful gains. Reallocate resources to higher-return activities before that point rather than pushing past it.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one area where you are investing heavily and ask whether the last 20% of effort is producing meaningful results.

YHK

Falsifiability

- A theory must be testable to be scientific.
- Philosopher Karl Popper's key principle.
- E.g., Astrology in India: not falsifiable = pseudoscience.
- Good economic models are those we can disprove.
- E.g., ISRO missions are evaluated on clear success/failure.
- Avoid vague goals or models that can't fail , they're useless.

YHK

Falsifiability

- **Definition**: A claim is only scientifically meaningful if it is possible, in principle, to prove it wrong, unfalsifiable beliefs are beyond the reach of evidence.
- **How to Apply**: For any belief or hypothesis, ask: "What evidence would convince me I am wrong?" If no such evidence exists, treat the belief as a matter of faith rather than fact.

(Ref: "Why trying to prove yourself wrong is the key to being right" - Skeptical Science)

Try This: Pick one strongly held belief and define the specific observation that would cause you to abandon it.

YHK

Survivorship Bias

- Focus on successes and ignore failures.
- WWII bomber example; key in statistics.
- E.g., highlighting startup unicorns, ignoring 90
- Bollywood stories of actors from small towns miss thousands who failed.
- E.g., glorifying IIT success without showing coaching burnout cases.
- Always ask: "What am I not seeing?"

YHK

Survivorship Bias

- **Definition**: We draw conclusions from visible successes while ignoring the invisible failures that never made it through the filter.
- **How to Apply**: Whenever you study success stories, actively ask what happened to those who tried the same approach and failed. Look for the graveyard of failed attempts before modelling your strategy on the survivors.

(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Find one success story you admire and research the failure rate of others who followed the same path.

YHK

The Peter Principle

- People get promoted to their level of incompetence.
- Formulated by Laurence J. Peter.
- Seen in bureaucracies and Indian PSUs.
- E.g., great engineer made manager , fails at leadership.
- Indian government often promotes seniority, not ability.
- Avoid automatic promotions , match role to strengths.

YHK

The Peter Principle

- **Definition**: In hierarchies, people are promoted based on performance in their current role until they reach a level of incompetence, where they stay.
- **How to Apply**: Promote people based on the skills required in the new role, not just their performance in the old one. Recognise that an excellent individual contributor may not be an effective manager.

"Your promotion follows the Peter Principle. When Peter likes you, you get promoted."

(Ref: Four Week MBA)

Try This: Evaluate your next promotion decision on the requirements of the destination role, not the track record in the current one.

YHK

Tragedy of the Commons

- Individuals overuse shared resources for personal gain.
- Coined by Garrett Hardin, 1968.
- E.g., overgrazing in Indian village pastures.
- Overfishing in coastal India, water tanker usage in Bangalore.
- Pollution of Yamuna due to collective neglect.
- Collective solutions + incentives required to escape the trap.

YHK

Tragedy of the Commons

- **Definition**: Shared resources are depleted when individuals, acting in their own rational self-interest, collectively over-consume them.
- **How to Apply**: Identify any shared resource, attention, budget, infrastructure, where individual incentives conflict with collective welfare. Design governance rules, quotas, or private ownership structures to align individual and group interests.

(Ref: Four Week MBA)

Try This: Spot one shared resource in your team or organisation that is being over-used and propose one constraint or governance rule.

YHK

Reciprocity

- People respond to kindness with kindness.
- From social psychology, basis of social contracts.
- E.g., Indian wedding invites $\rightarrow$ social obligation to reciprocate.
- Business offers: "free" samples $\rightarrow$ expectation to buy.
- Indian politics: favors given in return for votes.
- Can be manipulated , be conscious of emotional traps.

YHK

Reciprocity

- **Definition**: People feel a powerful obligation to return favours, gifts, and concessions, giving first creates a natural pull toward return.
- **How to Apply**: Build relationships by giving value first without expectation. In negotiation, make a concession and the other party will feel compelled to reciprocate. Be aware that reciprocity can also be weaponised through unsolicited gifts.

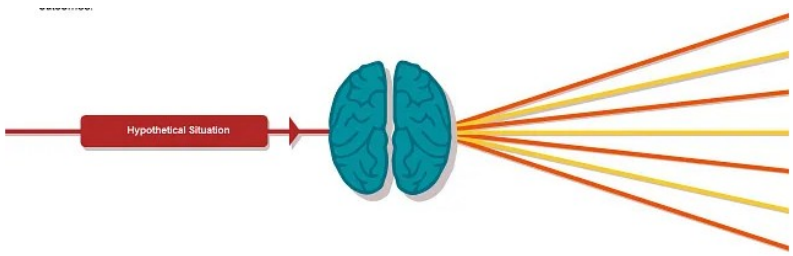
(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Offer genuine help to one person in your network this week with no immediate ask attached.

YHK

Thought Experiment

- Imagining a scenario to reason abstractly.
- Used by Einstein, Schrödinger, philosophers.
- E.g., "What if no one followed traffic signals in Delhi?"
- Indian planners simulate monsoon scenarios for disaster prep.
- Business leaders imagining customer journeys to refine UX.
- Avoid mental experiments with flawed premises.



(Ref: The Art Of Thinking Clearly - Swabhav Tech Labs)

Thought Experiment

- **Definition**: Explore hypothetical scenarios in the mind to test principles, stress-test beliefs, or generate insight without real-world experimentation.
- **How to Apply**: Construct a simplified imaginary scenario that isolates the key variable you want to understand. Reason through the scenario rigorously. Use the conclusion to inform your real-world view or decision.

(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Ask yourself: "What would happen if I doubled my price / halved my team / removed this feature?" and reason it through completely.

YHK

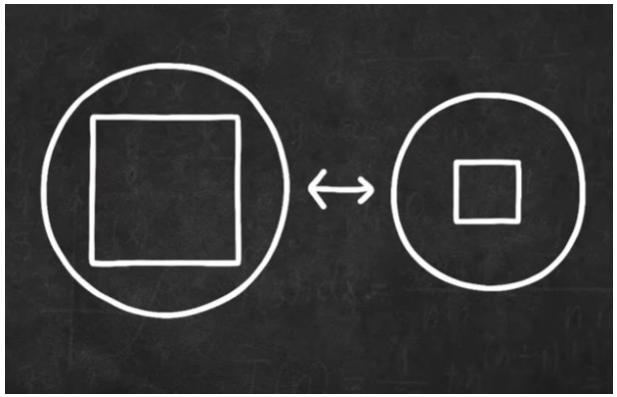
Relativity

- Perception depends on comparison.
- Not just physics , applies in psychology and economics.
- E.g., Rs.1000 seems big at a tea shop, small at Croma.
- Discounted MRPs manipulate perception of value.
- E.g., coaching institutes: "Only Rs.1 lakh" vs. "Just 2 months' salary."
- Avoid judging in isolation; watch your frame of reference.

YHK

Relativity

- **Definition**: Observations of speed, position, time, and many human judgments are not absolute, they depend entirely on the observer's frame of reference.
- **How to Apply**: When evaluating a situation, identify what frame of reference you are using and how it differs from others involved. Deliberately adopt another vantage point to test whether your conclusions hold.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: In a current disagreement, write a full description of the situation from the other party's perspective before responding.

YHK

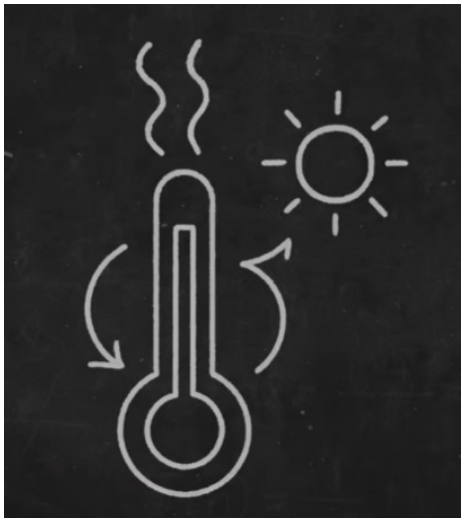
Thermodynamics

- Energy systems tend toward disorder (entropy).
- From physics; metaphor in organizational design.
- E.g., Indian government departments decay without active effort.
- Relationships degrade without energy (attention, care).
- E.g., neglected public infrastructure like Indian toilets, libraries.
- Systems must be fed with effort to counter entropy.

YHK

Thermodynamics

- **Definition**: Closed systems naturally drift toward disorder (entropy) over time; maintaining order requires continuous energy input.
- **How to Apply**: Apply this to organisations, relationships, and projects: without deliberate maintenance and investment, they degrade. Build regular "energy inputs", reviews, re-alignment sessions, maintenance cycles, into every system you run.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one system in your life or work that is drifting toward disorder and schedule a deliberate maintenance action.

YHK

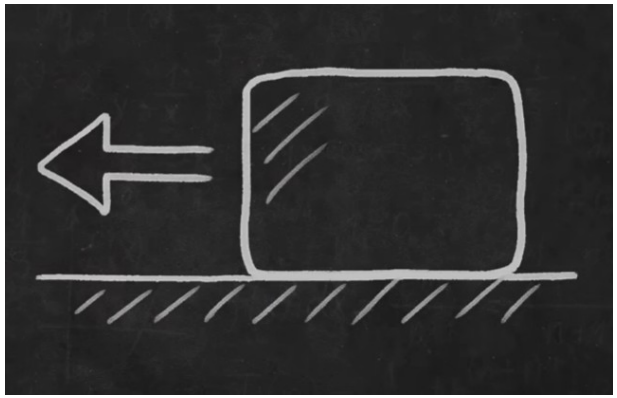
Inertia

- Objects at rest stay at rest; systems resist change.
- From Newtonian physics; metaphor for habits, culture.
- Bureaucracies in India resist reform , status quo bias.
- People stick to routines despite better alternatives.
- E.g., cash use persists despite UPI in rural India.
- Momentum can be good or bad , be mindful of direction.

YHK

Inertia

- **Definition**: Objects, and organisations, habits, and people, tend to remain in their current state until acted upon by a sufficient external force.
- **How to Apply**: To create change, apply enough concentrated force to overcome initial resistance rather than spreading effort thinly. To prevent unwanted change, build in stabilising structures that resist outside disruption.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one habit you want to change and design one specific, high-force trigger to overcome the inertia of the current state.

YHK

Friction and Viscosity

- Resistance slows systems down.
- Physics origin, applied in design thinking.
- E.g., paperwork in Indian passport process adds friction.
- Online education failing due to tech/infrastructure drag.
- E.g., MSME loans delayed by regulatory viscosity.
- Reduce friction strategically, but preserve checks where needed.

YHK

Friction and Viscosity

- **Definition**: Resistance embedded in a system, whether physical, procedural, or psychological, slows the flow of effort and output.
- **How to Apply**: Map the friction points in any process: unnecessary approvals, unclear steps, tool-switching costs. Eliminate or reduce each one. Conversely, add deliberate friction to behaviours you want to discourage.

(Ref: "Difference Between Friction and Viscosity" - Pediaa)

Try This: Count the steps in one recurring process and remove at least one step that adds no value.

YHK

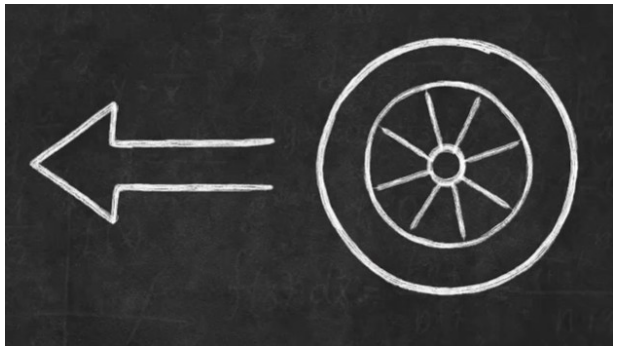
Velocity

- Speed in a particular direction.
- Physics model applied to business and policy.
- E.g., Indian startup growth = velocity, not just speed.
- Quick reforms + right direction = compounding effect.
- E.g., GST rollout aimed to increase fiscal velocity.
- Beware of high speed in wrong direction , causes more harm.

YHK

Velocity

- **Definition**: Unlike speed, velocity includes direction, moving fast in the wrong direction is worse than moving slowly in the right one.
- **How to Apply**: Define both the rate of progress and the direction before optimising for speed. Regularly confirm that your fast-moving efforts are still pointed at the right goal; adjust direction before accelerating further.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Ask of your current most urgent project: "Are we going fast in the right direction, or just going fast?"

YHK

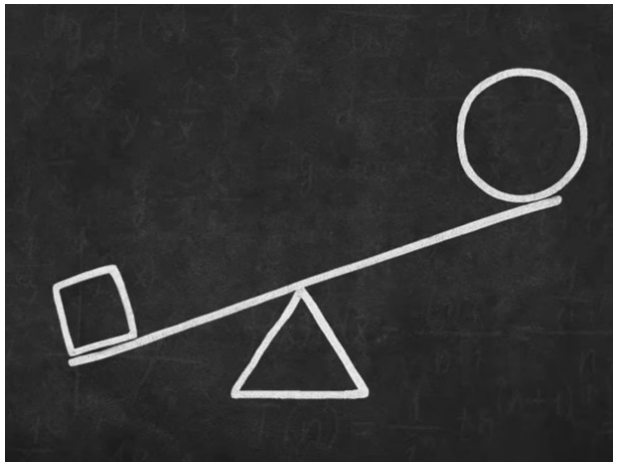
Leverage

- Small inputs yielding large outputs.
- Originates from mechanics and finance.
- E.g., using tech (like UPI) to scale access to banking.
- Influencers use platforms to gain massive reach.
- Debt in real estate , high leverage in Indian housing boom.
- Use leverage carefully; too much can break the system.

YHK

Leverage

- **Definition**: A small, well-placed force applied at the right point can move a load far beyond what direct effort alone could achieve.
- **How to Apply**: Find the highest-leverage point in any system, a key relationship, a bottleneck process, a critical assumption. Apply disproportionate effort or resource there rather than distributing it evenly.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: List your five current activities and mark the one that unlocks the most value for others, invest there first tomorrow.

YHK

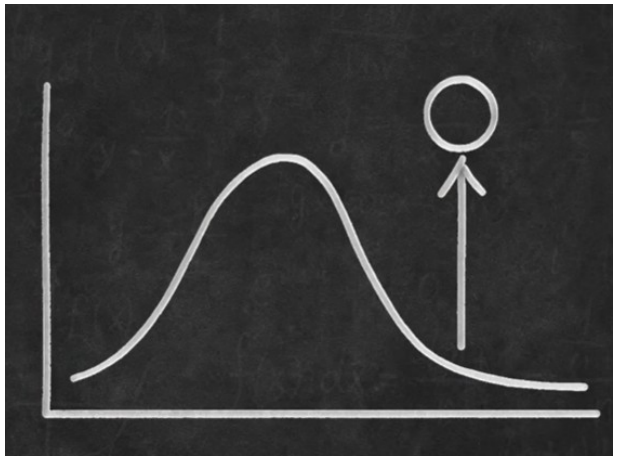
Activation Energy

- Minimum energy required to start a reaction.
- Chemistry origin; metaphor for habits and change.
- E.g., initial setup of solar in Indian homes is high, then payoff.
- Form-filling keeps citizens from accessing subsidies.
- E.g., waking up early needs energy "spike" at first.
- Lowering activation energy = increasing adoption.

YHK

Activation Energy

- **Definition**: Every process requires an initial energy input to get started, lower that barrier and the reaction proceeds naturally.
- **How to Apply**: Identify what makes starting hard, ambiguity, setup cost, emotional resistance, and reduce it. Make the first step trivially easy. In habit formation, design the environment so the activation cost of the desired behaviour is near zero.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Reduce the first step of one important habit to under two minutes so it requires almost no activation energy.

YHK

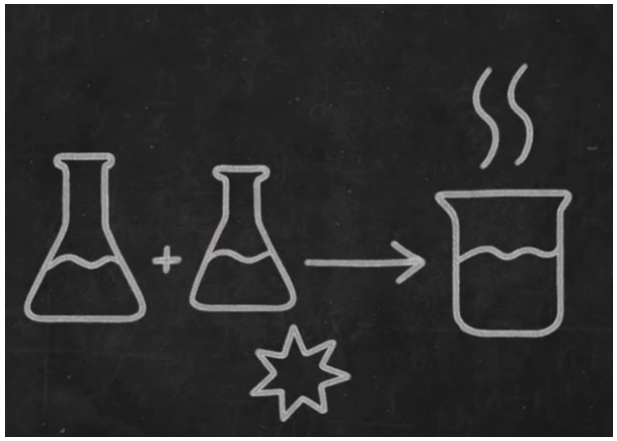
Catalysts

- Something that accelerates change without being consumed.
- From chemistry; used in transformation processes.
- E.g., a good mentor can catalyze career in Indian startups.
- Policies like Make in India as industrial catalysts.
- E.g., cricket coach who turns underdog into star.
- Catalysts don't guarantee success , they enable it.

YHK

Catalysts

- **Definition**: A catalyst accelerates a process without being consumed, it lowers the barrier to reaction without changing the start or end state.
- **How to Apply**: Look for people, tools, or conditions that accelerate progress without requiring proportional ongoing effort. Invest in catalytic elements, the right introduction, the right software, the right framing, that make everything else flow faster.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one bottleneck in a current project and ask: "What one person, tool, or change would make this much easier without me carrying it?"

YHK

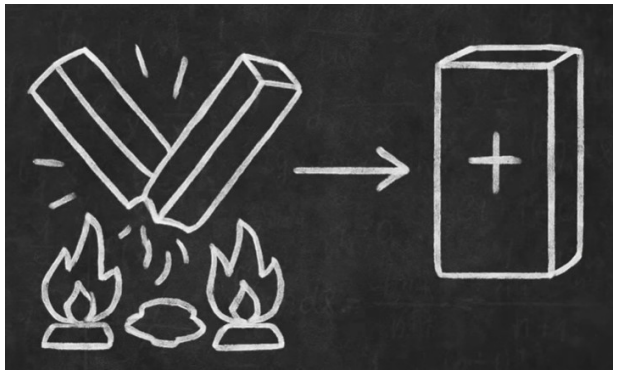
Alloying

- Combining elements for strength.
- From metallurgy; used metaphorically in teams/ideas.
- E.g., diverse coalition governments in India.
- Interdisciplinary education = alloy of skills.
- E.g., Hindi + tech = strong vernacular apps.
- Alloys may have trade-offs , balance strength vs. flexibility.

YHK

Alloying

- **Definition**: Combining two or more different materials or disciplines creates something stronger than any single element alone.
- **How to Apply**: Deliberately mix expertise, perspectives, or resources from different domains. The strongest teams, products, and ideas often emerge from cross-disciplinary combinations that create properties neither element possesses individually.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Bring one person from a completely different field into your next brainstorm and listen for how their framing changes your thinking.

YHK

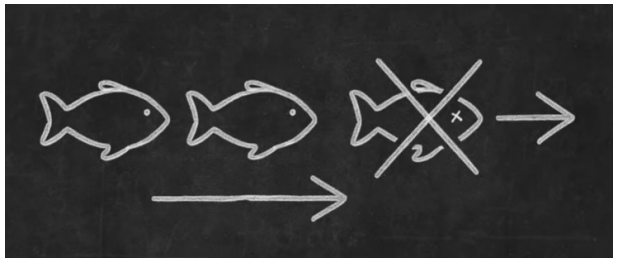
Evolution - Natural Selection

- Survival of the fittest over generations.
- Darwinian principle; applicable to markets.
- Indian startups that adapt thrive (Zerodha, Paytm).
- Weak policies die out over time; strong ones stay.
- E.g., traditional farming surviving despite modern competition.
- Avoid romanticizing all survivors , some survive by chance.

YHK

Evolution, Natural Selection

- **Definition**: Variation plus differential survival equals adaptation, entities best suited to their environment reproduce and persist while others fade.
- **How to Apply**: Run many small experiments (variation), measure results honestly (selection), and double down on what survives. Apply this to products, strategies, and habits: create options, test, keep what thrives, discard what doesn't.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Run three small, low-cost experiments on a current challenge this month and kill the two that perform worst.

YHK

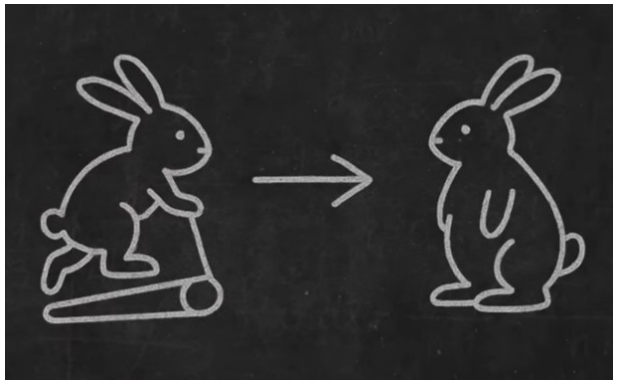
Evolution - Adaptation

- Organisms change to fit environment.
- Evolutionary biology principle.
- E.g., Indian retail moving from cash to QR codes.
- COVID forced remote working and online schooling.
- Small businesses adopted e-commerce or died.
- Adaptation is not always optimal , may be reactive.

YHK

Evolution, Adaptation

- **Definition**: Organisms change over generations to better fit their environment, the features that aid survival are retained and amplified.
- **How to Apply**: Build adaptive capacity into systems and teams. Review periodically whether your current approach still fits the environment, and redesign proactively rather than waiting for crisis to force change.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Review one process or habit that hasn't changed in over a year and ask whether the environment it was designed for still exists.

YHK

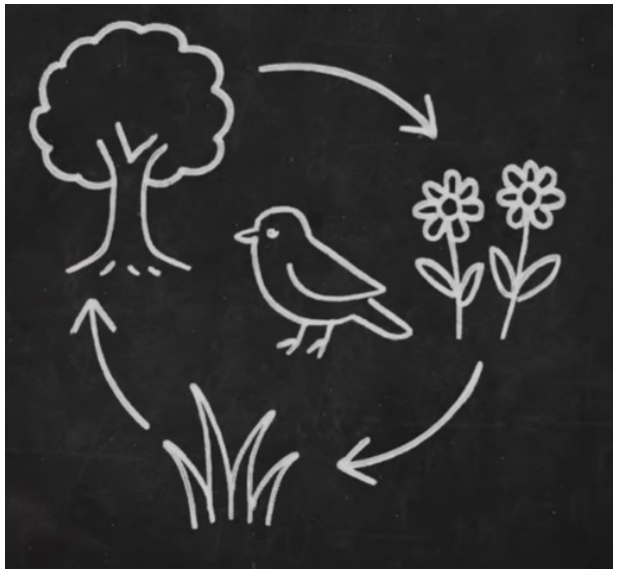
Ecosystems

- Interdependent networks of entities.
- From ecology; applied to business, tech.
- E.g., India's startup ecosystem: VCs, colleges, infra.
- Agriculture involves weather, soil, politics, supply chain.
- Bollywood = actors + OTT + media + fans.
- Don't isolate systems , analyze interactions.

YHK

Ecosystems

- **Definition**: A community of interdependent elements, organisms, organisations, or ideas, that sustain each other through flows of energy, information, or value.
- **How to Apply**: Map the ecosystem around your product, team, or community: who produces, who consumes, who decomposes and recycles. Design for the health of the whole system, not just your node within it.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Draw a simple map of five entities that your work depends on and identify the most fragile link in that web.

YHK

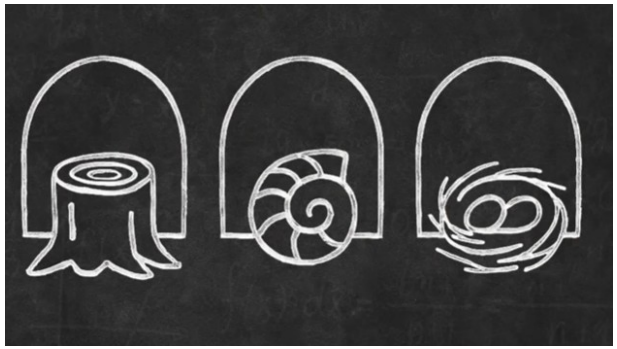
Niches

- Specialized roles within systems.
- From biology, used in marketing and strategy.
- E.g., Indian vernacular YouTubers targeting Tier-2 cities.
- Brands like Patanjali tapping Ayurvedic niche.
- Job roles like drone pilot, data annotator = new niches.
- Niches can be fragile , depend on broader system health.

YHK

Niches

- **Definition**: In any ecosystem, lasting success comes from finding and occupying a distinct role that others are not filling, not from competing head-on for the same resources.
- **How to Apply**: Define the specific, narrow problem you solve better than anyone else. Resist pressure to compete on every dimension. Depth in a niche creates durable advantage that broad competition erodes.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Write a one-sentence description of the precise niche your work occupies, if it sounds like everyone else, sharpen it.

YHK

Self-Preservation

- Tendency of organisms/systems to protect themselves.
- Evolutionary instinct; metaphor in institutions.
- E.g., Indian babus resisting administrative reforms.
- People resist feedback that threatens ego.
- E.g., companies killing innovations to protect core business.
- Self-preservation is natural, but can block growth.

YHK

Self-Preservation

- **Definition**: Every organism, system has an instinct to protect its own existence, this drive shapes behaviour even when it conflicts with growth/truth.
- **How to Apply**: Recognise self-preservation instincts. When resistance to change feels strong, ask whether it serves genuine wellbeing or merely protects ego, status, or comfort. Create safe conditions for change to reduce the threat response.

(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one area where you or your team resist feedback, and ask honestly whether that resistance protects quality or just comfort.

YHK

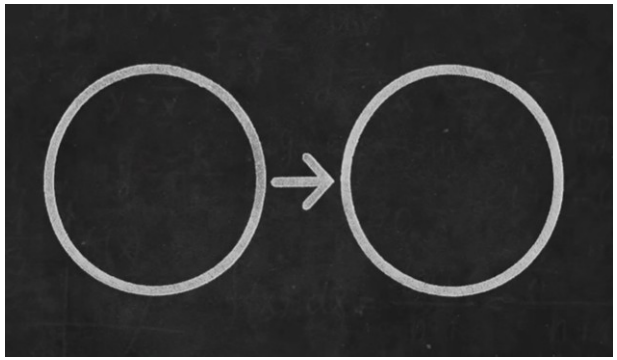
Replication

- Ability to copy and scale reliably.
- From biology and manufacturing.
- E.g., Indian franchise model , CCD, Amul parlors.
- Education: coaching classes cloned in every city.
- E.g., political strategies replicated across states.
- Mindless replication can ignore local context.

YHK

Replication

- **Definition**: The ability to copy a successful pattern faithfully, and the recognition that imperfect copying introduces variation that can drive improvement.
- **How to Apply**: Document successful processes so they can be replicated consistently. Allow enough variation in replication to surface improvements. Distinguish between errors (fix them) and productive mutations (explore them).



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Document one process you do repeatedly so precisely that someone else could replicate it with zero guidance from you.

YHK

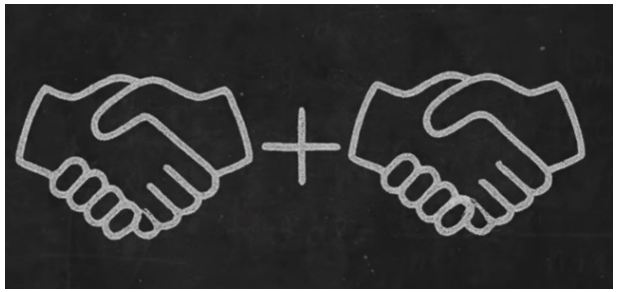
Cooperation

- Working together for mutual benefit.
- Core to human evolution and society.
- SHGs in India empowering rural women.
- Farmers pooling land/machinery.
- Swachh Bharat relied on mass cooperation.
- Needs trust, alignment , forced cooperation fails.

YHK

Cooperation

- **Definition**: Entities working together toward shared goals achieve outcomes that none could accomplish alone, the foundation of social organisation and complex systems.
- **How to Apply**: Design systems where cooperation is the rational choice. Align individual incentives with collective outcomes. Build trust through transparency and reciprocity. Make defection costly and collaboration rewarding.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Identify one recurring conflict in your team or organisation and redesign the incentive structure so that cooperation benefits everyone more than competition.

YHK

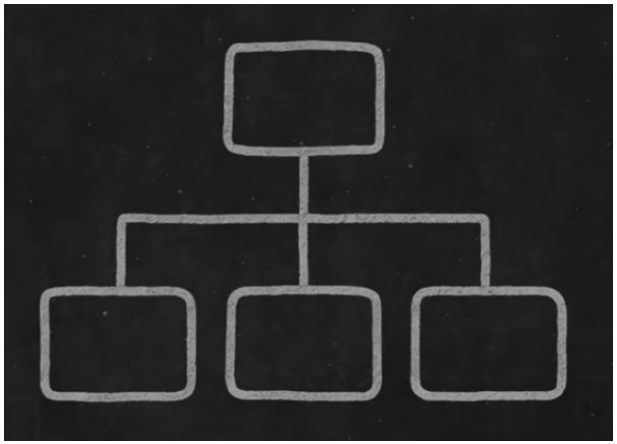
Hierarchical Organization

- Systems structured in layers of control.
- Found in biology, military, governance.
- E.g., Indian bureaucracy: central, state, district levels.
- Companies with vertical vs. flat hierarchies.
- Indian family structures: elders $\rightarrow$ juniors.
- Hierarchy brings order, but can stifle speed and creativity.

YHK

Hierarchical Organisation

- **Definition**: Complex systems organise into layers where higher levels coordinate lower ones, enabling scalability and specialisation at the cost of flexibility.
- **How to Apply**: Match hierarchy depth to the complexity and stability of the task. Use flatter structures for creative, fast-changing work; deeper hierarchies for large-scale execution. Ensure communication flows clearly across levels.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Map the decision hierarchy in your team and identify one decision currently made at a level too high or too low for speed and quality.

YHK

Incentives

- Behavior is shaped by rewards and penalties.
- Economics principle; vital in policy and business.
- E.g., LPG subsidy linked to Aadhaar = better targeting.
- Swiggy/Zomato drivers incentivized on deliveries.
- Voters swayed by freebie promises.
- Misaligned incentives = corruption, manipulation.

YHK

Incentives

- **Definition**: Behaviour follows rewards and penalties, people and systems reliably do more of what is rewarded and less of what is penalised.
- **How to Apply**: Before designing any system or policy, ask: "What behaviour does this actually reward?" Align the incentive structure with the outcome you genuinely want. Watch for unintended incentives that produce unwanted behaviour.

(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Examine one rule or process in your work and trace what behaviour it actually incentivises, then ask whether that matches the intended goal.

YHK

Tendency to Minimize Energy

- Systems prefer the path of least resistance.
- Physics + human psychology.
- E.g., people choose WhatsApp forwards over research.
- Politicians use slogans instead of nuanced solutions.
- E.g., shortcuts in JEE prep , coaching hacks.
- Shortcuts are tempting , but costly if misused.

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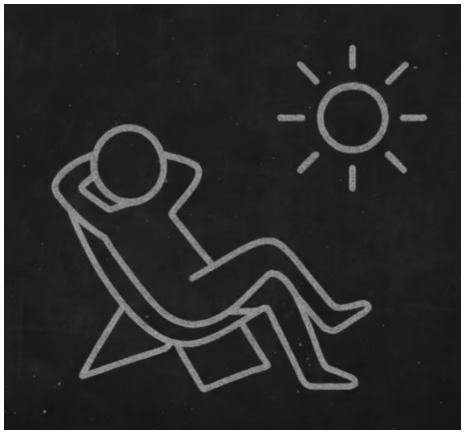
Tendency to Minimise Energy

- **Definition**: Physical and biological systems, and humans, naturally gravitate toward the path of least resistance, minimising energy expenditure wherever possible.
- **How to Apply**: Use this principle to design better systems: make the desirable behaviour the easiest option. In personal productivity, reduce the effort required for important habits and increase the friction of undesirable ones.



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Redesign your workspace or workflow so that your most important task is the lowest-resistance activity at the start of your day.

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Feedback Loops

- Outputs become new inputs, amplifying or correcting.
- From cybernetics; applies in economics, learning.
- E.g., social media likes $\rightarrow$ more posts $\rightarrow$ more likes.
- UPI usage loop: adoption $\rightarrow$ trust $\rightarrow$ more users.
- Crime in areas $\rightarrow$ police presence $\rightarrow$ less crime.
- Watch for both positive (amplifying) and negative (balancing) loops.

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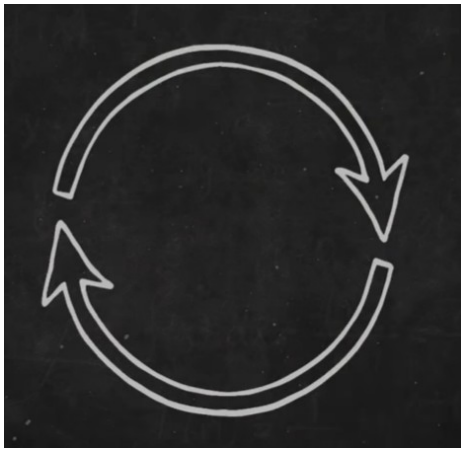
Feedback Loops

- **Definition**: A feedback loop occurs when the output of a system is fed back as input, either amplifying the original signal (reinforcing loop) or counteracting it (balancing loop).
- **How to Apply**: Identify the feedback loops operating in any system you manage. Strengthen reinforcing loops that drive desired outcomes; design balancing loops to prevent runaway effects. Shorten feedback cycles to accelerate



(Ref: "100 Most Powerful Mental Models to Fall Asleep to" - SleepWise)

Try This: Map one feedback loop in your daily routine or work, trace how the output of an action feeds back to influence your next action.

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Building Your Personal Mental Models

(Ref: A Comprehensive Guide to Mental Models - Habits for Thinking)

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Why Build a Latticework of Mental Models?

- Latticework means interlocking structure of big ideas from different disciplines
- Repeating mistakes? You're not alone.
- Best thinkers don't think harder,they think in models.
- A latticework = multiple models from various fields.
- Inspired by Charlie Munger's approach to decision-making.
- Models create clarity, prevent bias, and improve reasoning.

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The Brain as a Workshop

- Every mental model = a tool.
- Relying on one tool = repeated errors.
- Full toolkit = deeper insight + fewer blindspots.
- Munger: "Array experience on a latticework of models."
- Real decisions require multiple perspectives.

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Why Interconnected Models Matter

- Complex problems need multi-model solutions.
- Examples:
  - Inversion (think backwards to prevent failure)
  - 80/20 Rule (focus on leverage points)
  - Second-order thinking (anticipate consequences)
- Together: see around corners, think systemically.

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Step 1: Cultivate Curiosity

- Read across disciplines: psychology, biology, economics, etc.
- Look for timeless, transferable ideas.
- Ask: "What's the principle here?"
- Apply ideas outside their original context.
- One book outside your comfort zone can shift your thinking.

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Step 2: Actively Understand Models

- Move from passive to active learning.
- Use the Feynman Technique: explain in plain language.
- Create analogies to what you know.
- Journal 3 uses per model.
- Seek cross-discipline examples for each model.

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Step 3: Find Model Connections

- Insight = when models intersect.
- Example intersections:
  - Feedback loops + habit formation
  - Marginal utility + diminishing returns
  - Hanlon's Razor + Occam's Razor
- Use mind maps or journals to visualize links.
- Practice with real decisions: apply 3 models at once.

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Step 4: Practice & Apply

- Use models like lenses.
- Ask: "What model fits this problem?"
- Try on everyday decisions:
  - Weekly planning → Pareto Principle
  - Disagreements → Steelman Technique
  - Scheduling → Parkinson's Law
- Repetition builds intuition.

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Step 5: Review and Refine

- Every quarter, reflect:
  - Which models do I use most?
  - Which still confuse me?
  - What should I add next?
- Keep a Mental Models Journal.
- Be willing to discard outdated models.
- Your latticework evolves with experience.

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Overcoming Common Challenges

- **Info Overload:** Start with 10 core models.
- **Time Constraints:** Use daily touchpoints,podcasts, short reads.
- **Shallow Learning:** Go deep, not wide.
- Value comes from mastery, not memory.

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The Lifelong Edge

- Your brain is a framework builder, not a sponge.
- A strong latticework lets you:
  - Make better decisions
  - Spot unseen patterns
  - Avoid mental traps
  - Communicate clearly
- This is lifelong work. Build with care.
- Your mental toolkit is your edge.

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Closure

Key Takeaways

- **Think in models, not opinions**, models force you to reason, not just react.
- **No single model is enough**, use a latticework; different lenses reveal different truths.
- **Models are tools, not truths**, always update when reality contradicts your map.
- **The best models are cross-disciplinary**, borrow from physics, biology, economics, psychology.
- **Practice daily**, apply one model to one real decision every day; mastery follows.
- **Beware of biases and fallacies**, knowing a model is not the same as being immune to its opposite failure mode.

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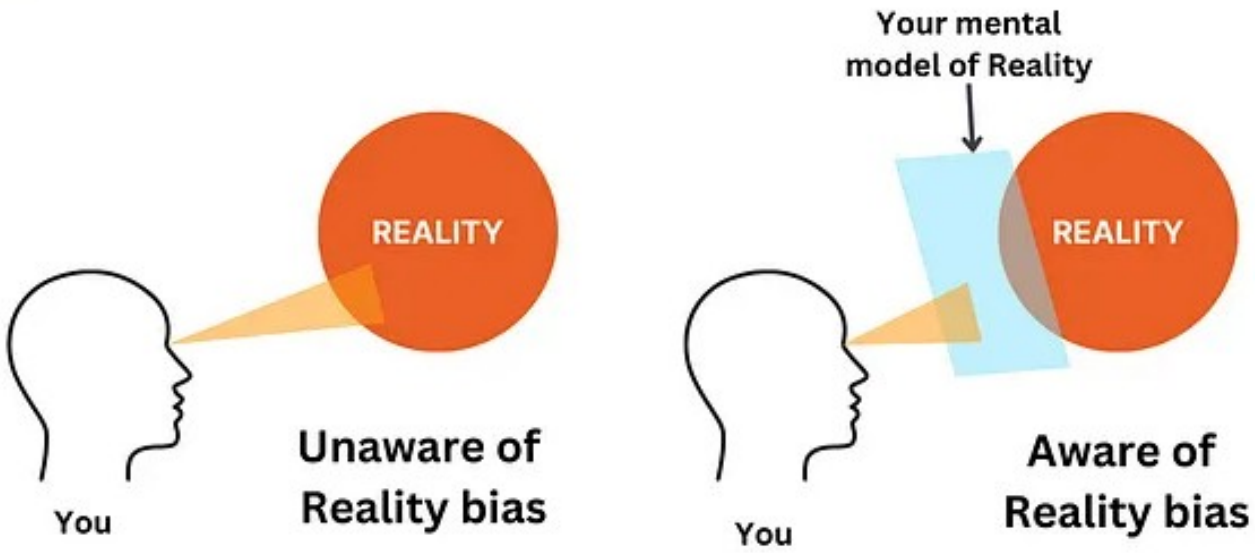
Closure

- A mental model is simply a representation of how something works.
- We cannot keep all of the details of the world in our brains, so we use mental models to simplify the complex into understandable and organizable chunks.



(Ref: The Art Of Thinking Clearly - Swabhav Tech Labs)

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Thanks ...

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